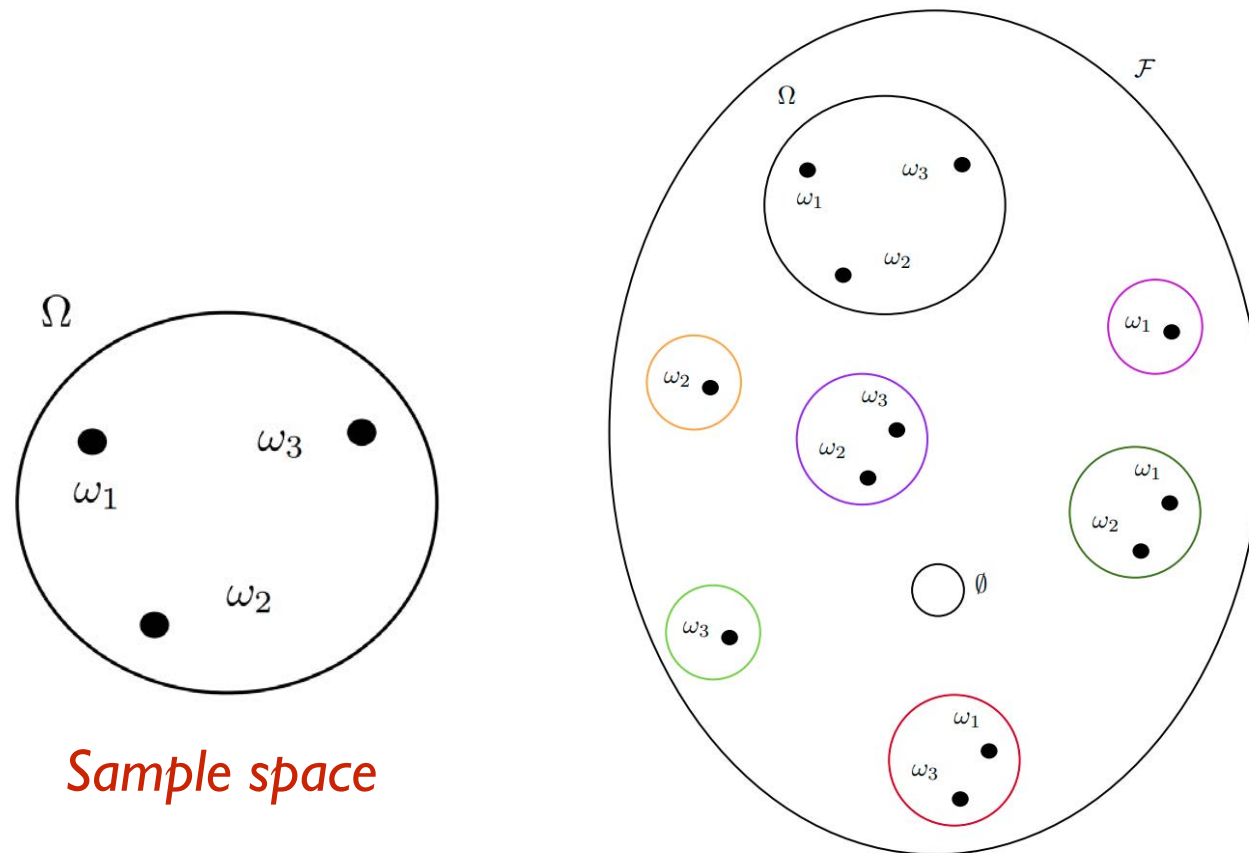


ENM 5310: Data-driven Modeling and Probabilistic Scientific Computing

Lecture #2: Primer on Probability

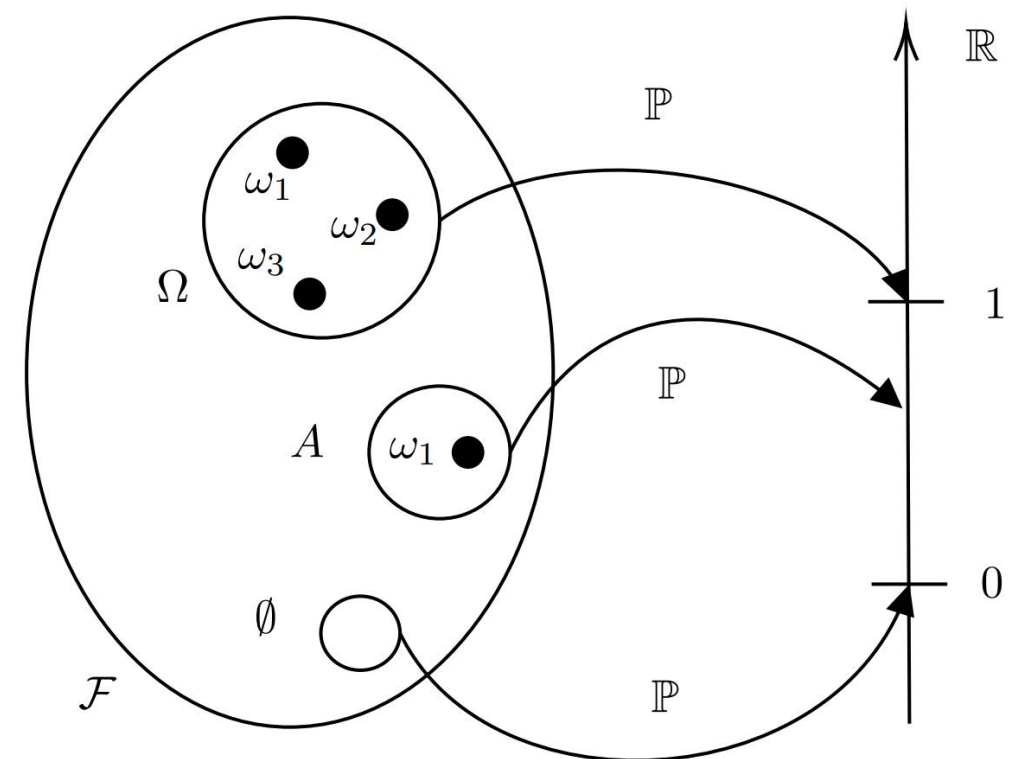


Probability spaces & random variables



Sample space

σ -algebra of events



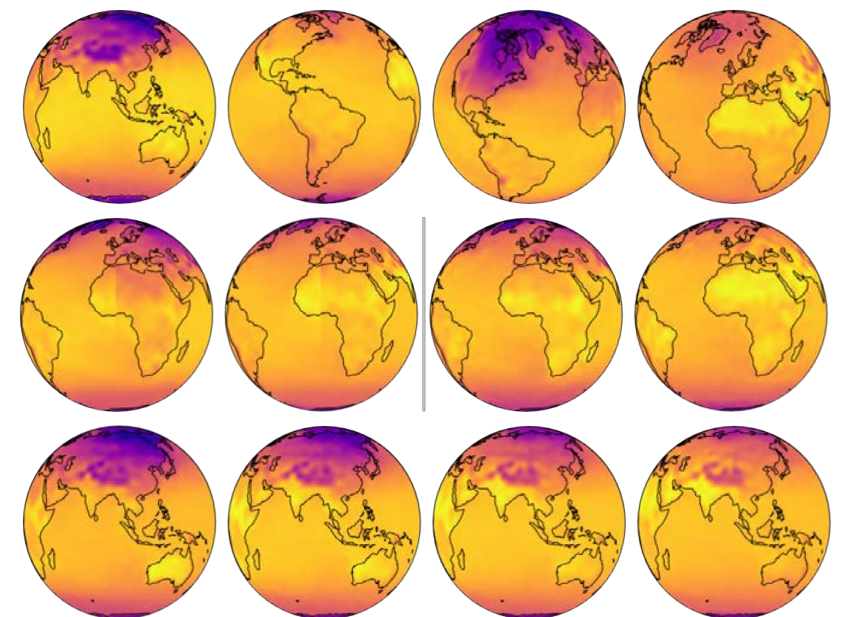
Probability measure



vectors



matrices



functions

Recap: Continuous random variables

- A ***continuous random variable*** is one which takes an infinite number of possible values. Continuous random variables are usually measurements. Examples include height, weight, the amount of sugar in an orange, the time required to run a mile.
- A continuous random variable is not defined at specific values. Instead, it is defined over an *interval* of values, and is represented by the ***area under a curve*** (in advanced mathematics, this is known as an *integral*). The probability of observing any single value is equal to 0, since the number of values which may be assumed by the random variable is infinite.

Probability density functions

$$\Pr[a \leq X \leq b] = \int_a^b f_X(x) dx.$$

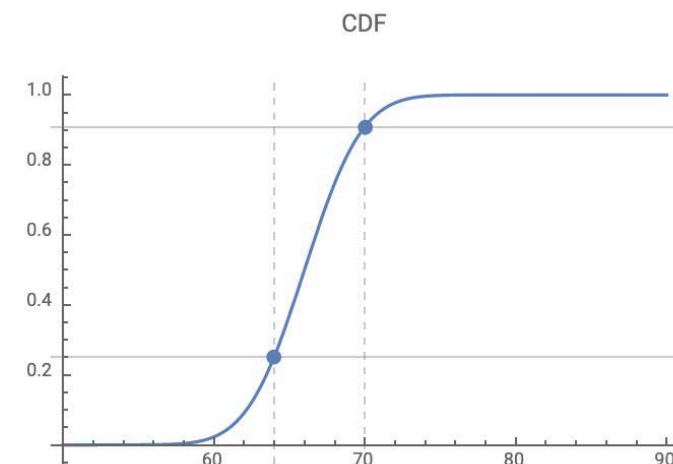
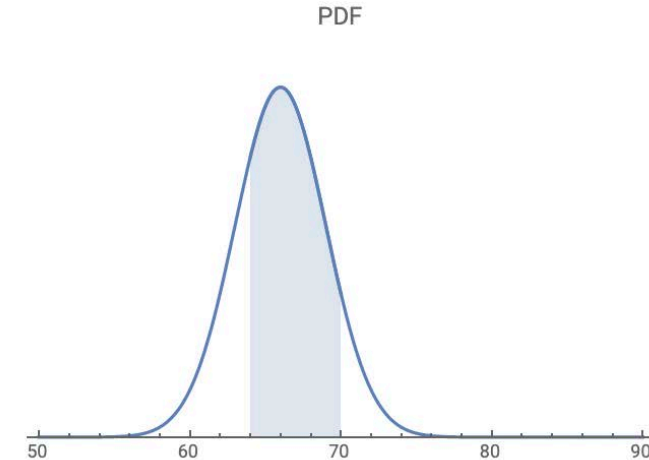
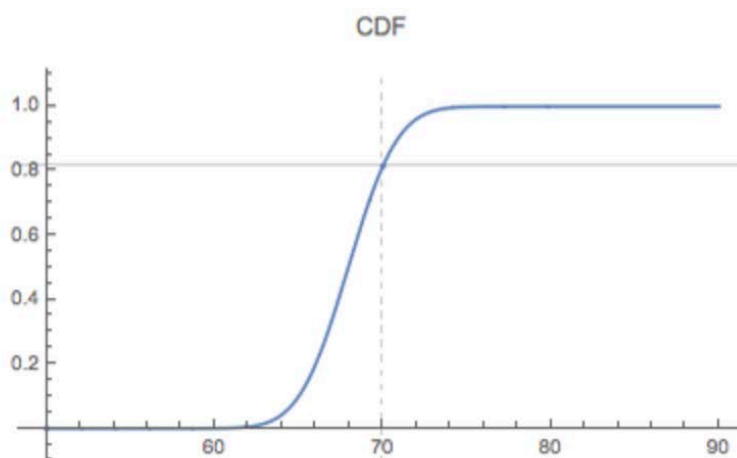
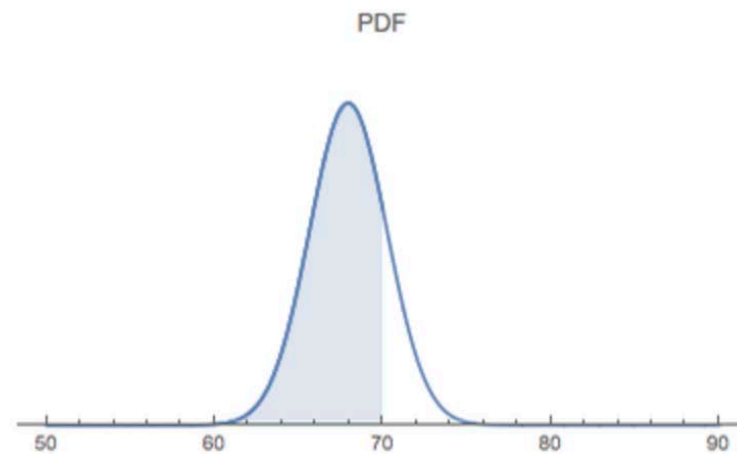
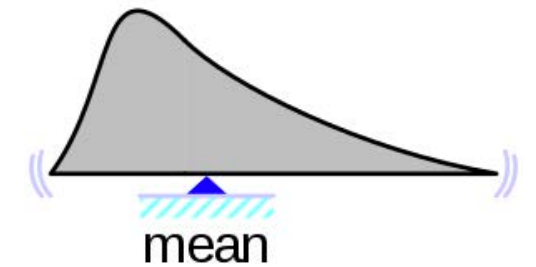
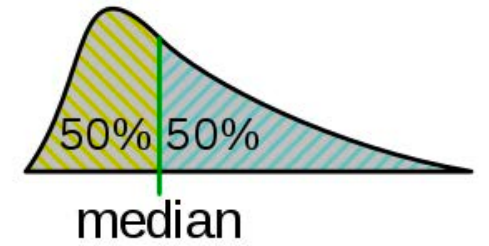
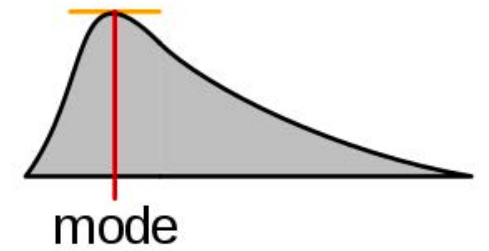
Hence, if F_X is the **cumulative distribution function** of X , then:

$$F_X(x) = \int_{-\infty}^x f_X(u) du,$$

and (if f_X is continuous at x)

$$f_X(x) = \frac{d}{dx} F_X(x).$$

Intuitively, one can think of $f_X(x) dx$ as being the probability of X falling within the infinitesimal **interval** $[x, x + dx]$.



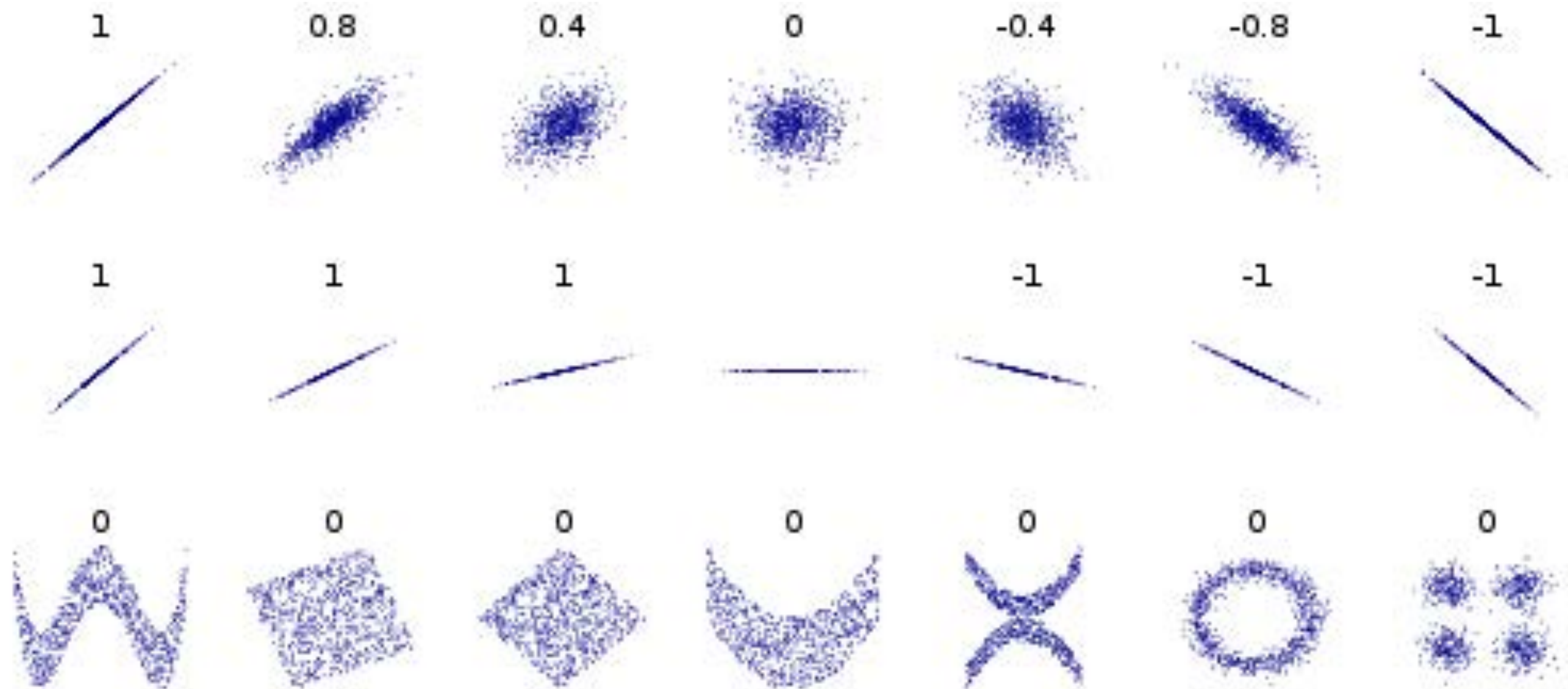
Basic rules of probability

Sum rule $p(X) = \sum_Y p(X, Y)$

Product rule $p(X, Y) = p(Y|X)p(X)$

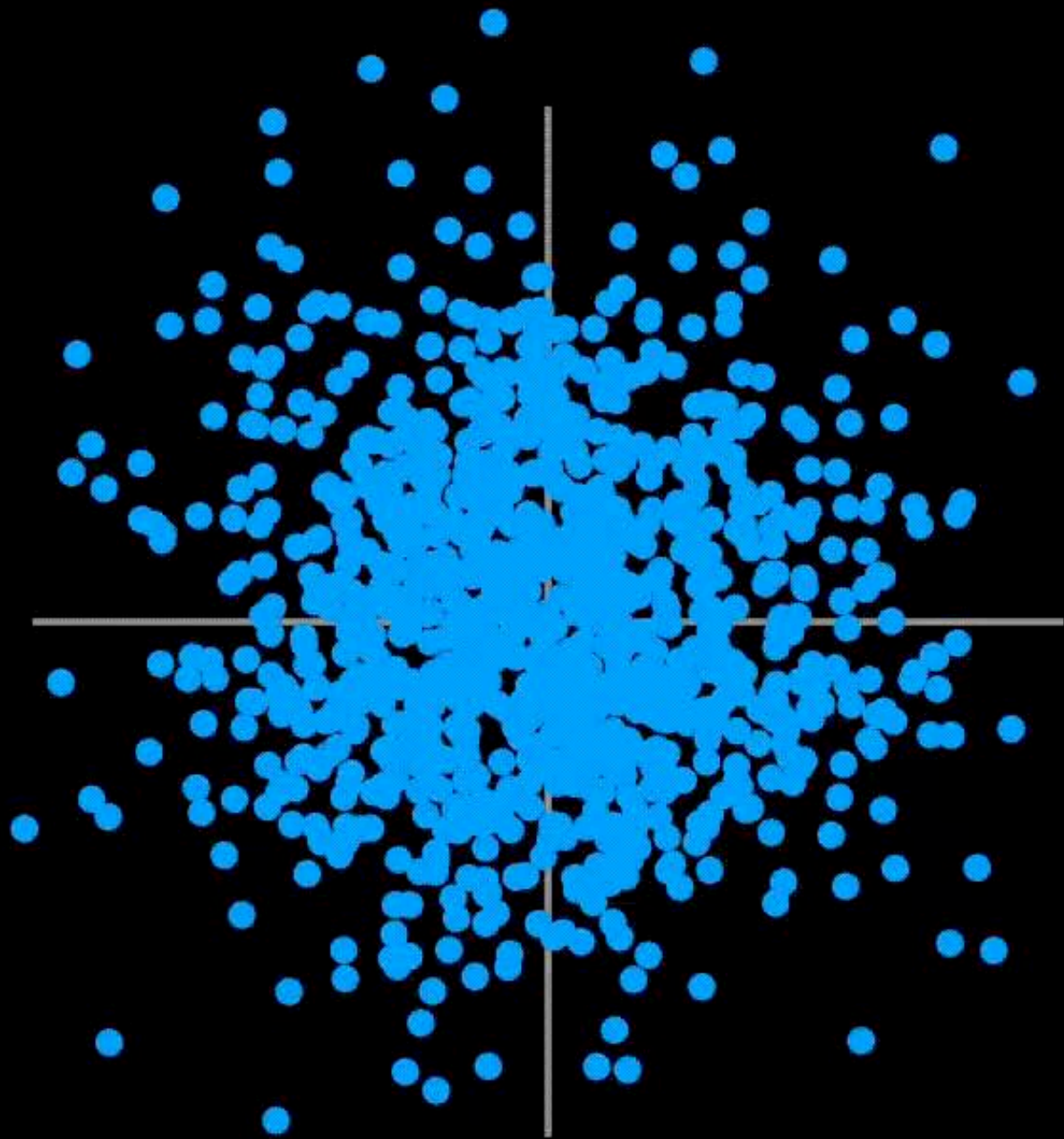
Bayes rule $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$

Correlation and linear dependence



Covariance vs Mutual Information

$$\text{cov}(X, Y) \quad I(X; Y)$$



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